TAHSIN NAYEEM SHRESTHA

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EDUCATION

Stony Brook University, *New York, NY, USA.* *05/2026*
Bachelor of Science, Information Systems Engineering
Minor: Studio Arts
Specialization: Financial System Management
Honors and Awards: Women in Science & Engineering Program (WISE), Dean's List 2025, Dean's List 2023
Cornell University, *New York, NY, USA.* *05/2023 - 05/2024*
Certificate in Machine Learning Foundations and Artificial Intelligence

EXPERIENCE

SBU IT Department, Software Developer, *New York, NY, USA.* *09/2022 - 05/2026*
- Engineered an LLM-powered (GPT-4, Claude) ticket analysis pipeline processing 50,000+ TDX IT service records across multiple academic years, automating classification and summarization workflows that significantly reduced manual categorization effort and improved reporting turnaround.
- Developed a multi-stage LLM workflow using GPT-4 and Claude to triage unstructured ticket data, detect anomalies, and identify macOS/Windows imaging trends for campus IT operations.
- Partnered with a cross-functional data analytics team to design standardized reporting methodologies and PostgreSQL data quality pipelines, translating raw IT service data into actionable insights that directly supported data-driven decision-making at scale.

KPMG International Limited, AI Engineer Apprentice, *New York, NY, USA.* *07/ 2023 - 12/2023*
- Spearheaded a Python-based analysis of 1M+ U.S. Census 2020 records stored in AWS S3, building and deploying regression, ensemble, and neural network models that quantified income disparities in New York State directly informing equity policy recommendations.
- Engineered predictive modeling pipelines with NumPy, Pandas, and Seaborn, using DynamoDB for scalable record management to detect gender-based salary gaps of up to 15% in low-wage sectors, improving model interpretability and decision-support accuracy by 20%.
- Generated model-driven insights supporting equity analysis through iterative model evaluation and benchmarking.

Division of Information and Technology, Technology Assistant, *New York, NY, USA.* *09/2022 - 05/2026*
- Delivered technical support to 500+ students, faculty, and staff via walk-ins, remote sessions, and on-site troubleshooting across 10+ campus buildings, ensuring 99% service satisfaction and minimal down-time.
- Performed 200+ hardware/software maintenance tasks, including OS imaging, secure data wiping, driver configuration, and device deployment; contributed to maintaining campus-wide print systems with 100% uptime.
- Trained and mentored 5+ new technicians, improving onboarding efficiency by 30% while streamlining documentation and ticket-resolution workflows within the IT service management platform.

PROJECTS

ScaleForge | Distributed Systems Simulation Platform *06/2026*
- Designed and built an interactive distributed systems simulator modeling request routing across load balancers, CDNs, caches, databases, and microservices.
- Implemented an event-driven architecture using a publish-subscribe event bus with real-time 3D visualization, distributed tracing, and live performance metrics.
- Developed and deployed a globally accessible application on Cloudflare Pages using native ES modules, emphasizing modular architecture, scalability, and maintainability.

Edge-Based Developer Feedback Analytics Platform *01/2026*
- Built a serverless analytics platform on Cloudflare Workers to aggregate developer collaboration data through REST APIs and low-latency edge processing.
- Developed REST API integrations and data pipelines in JavaScript, designing a scalable architecture for low-latency data processing without traditional servers.

Network Traffic & Protocol Analysis Toolkit *12/2025*
- Developed Python tools to analyze PCAP traffic, extracting flow-level metrics and evaluating TCP/DNS network performance through latency, throughput, packet loss, and congestion analysis.
- Applied systems-level debugging to interpret real-world network behavior and validate protocol performance assumptions.

SKILLS

Technical Skills: Python, JavaScript (ES6+), Java, SQL, C, Bash, React, Node.js, Django, PostgreSQL, SQLite, AWS (Lambda, API Gateway, S3, OpenSearch), Azure, Cloudflare Workers, TensorFlow, scikit-learn, Pandas, NumPy, Git, REST APIs, CI/CD, Distributed Systems, Microservices.